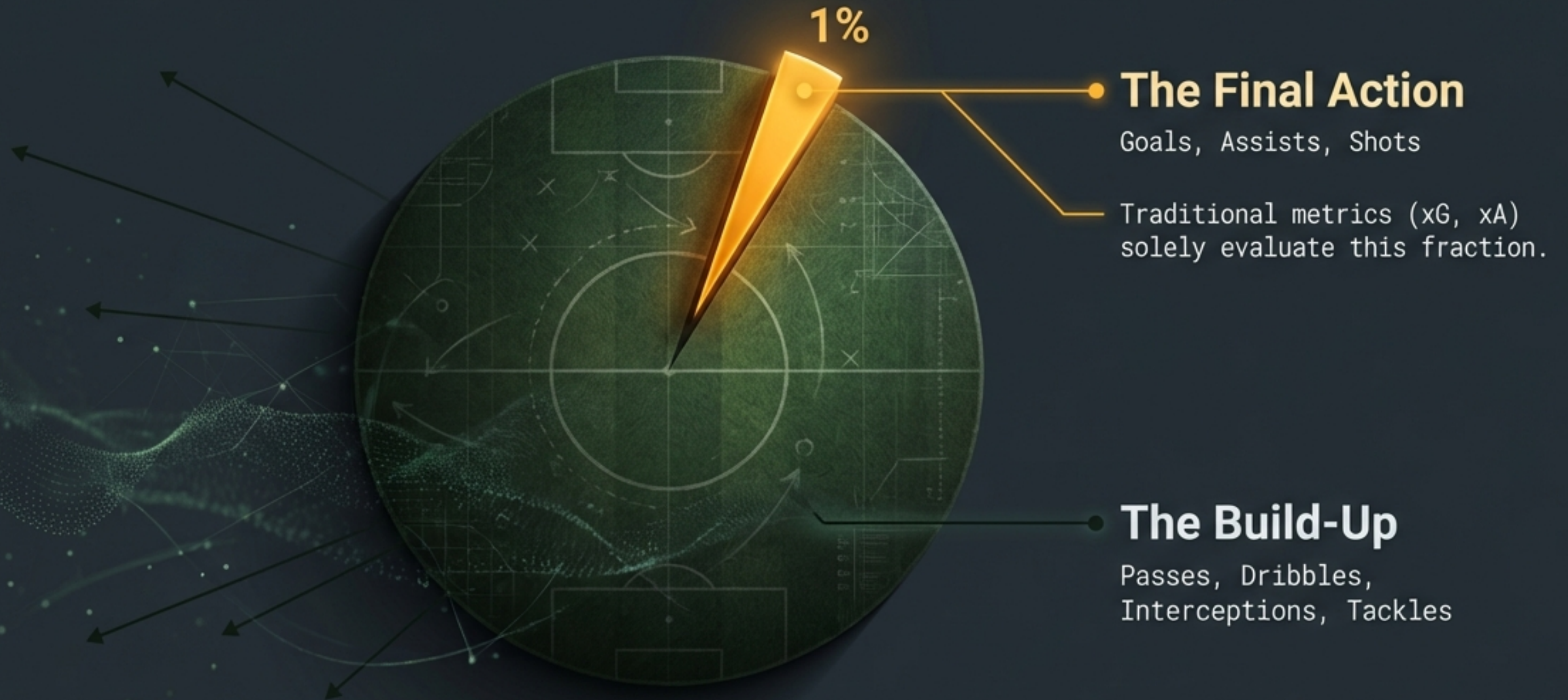


Valuing the 99 Percent of Football

The Evolutionary Arc of Possession State Models
and the Next Analytical Frontier



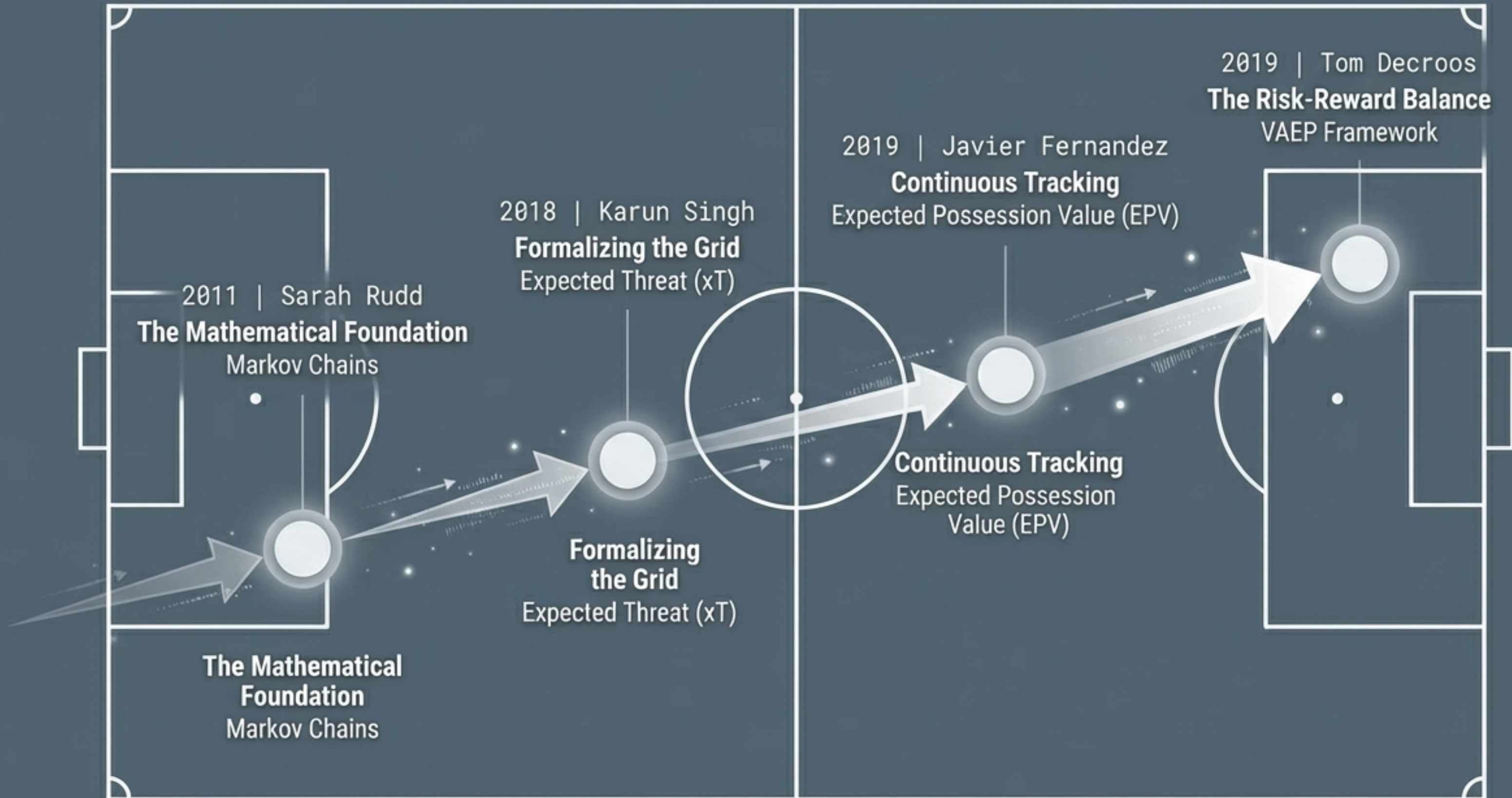
The Dark Matter of Football Analytics



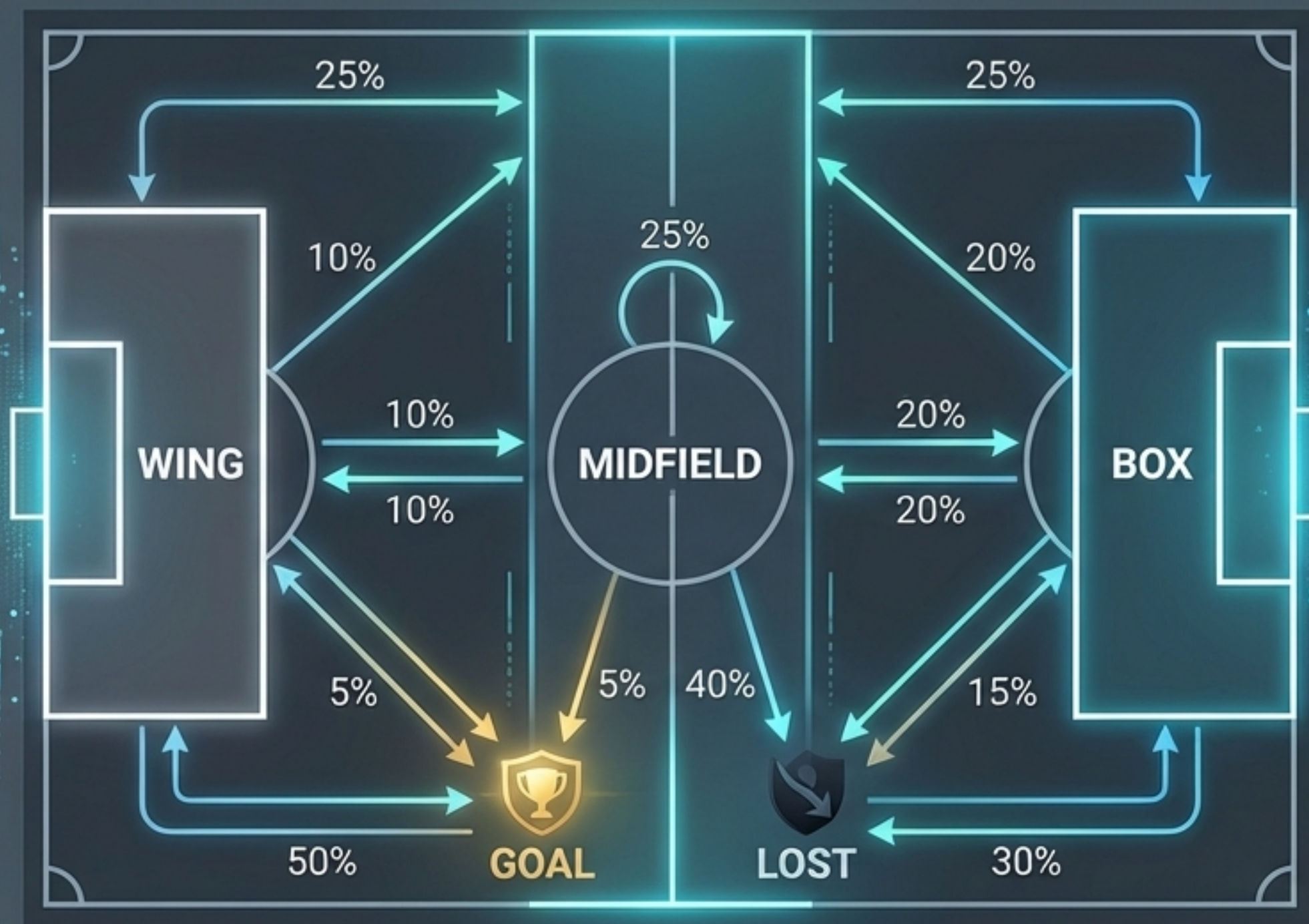
How do we quantify the **value of every single action** that moves a team closer to scoring?

The analytical solution is the development of **Possession State Value Frameworks**, which map the shifting probability of a goal at any given moment.

The Progression of State Value Models



Modeling Football as a Stochastic Process



 **Researcher:** Sarah Rudd (2011)

Innovation

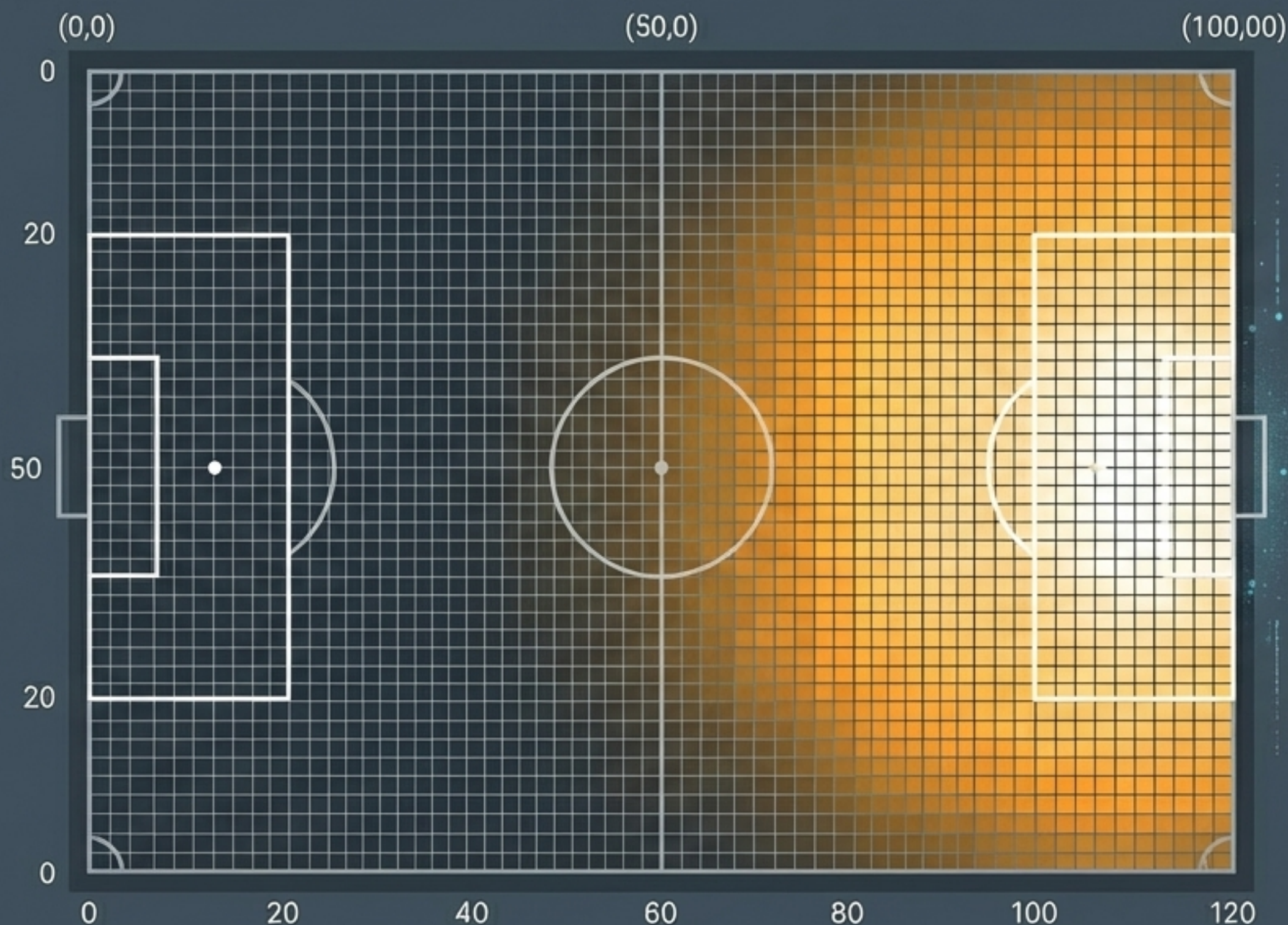
Shifted analytical focus from the isolated action to the broader **game state**.

Modeled possessions as a **Markov Chain** to calculate the long-term value of any position on the pitch.

Limitation

Constrained by **early event** data; relied on heavily simplified spatial zones.

Granular Valuation Through the Expected Threat Grid



$$xT = \text{Shot Threat} + \text{Move Threat}$$

 **Researcher:** Karun Singh (2018)

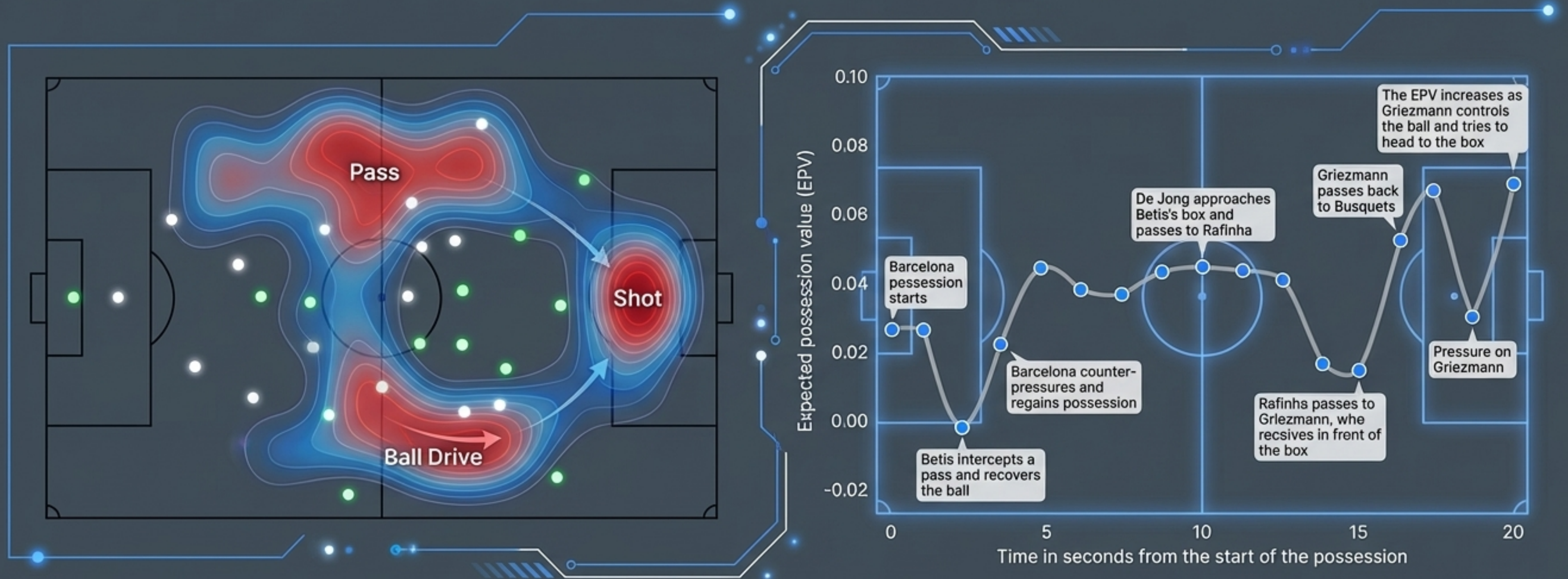
Innovation

Popularized the grid-based approach. Actions are valued purely by calculating the difference in xT between the start and end coordinates of a pass or carry.

Limitation

Only considers the ball's location. Ignores defensive pressure, off-ball context, and the risk of losing possession.

Integrating Spatiotemporal Tracking Dynamics



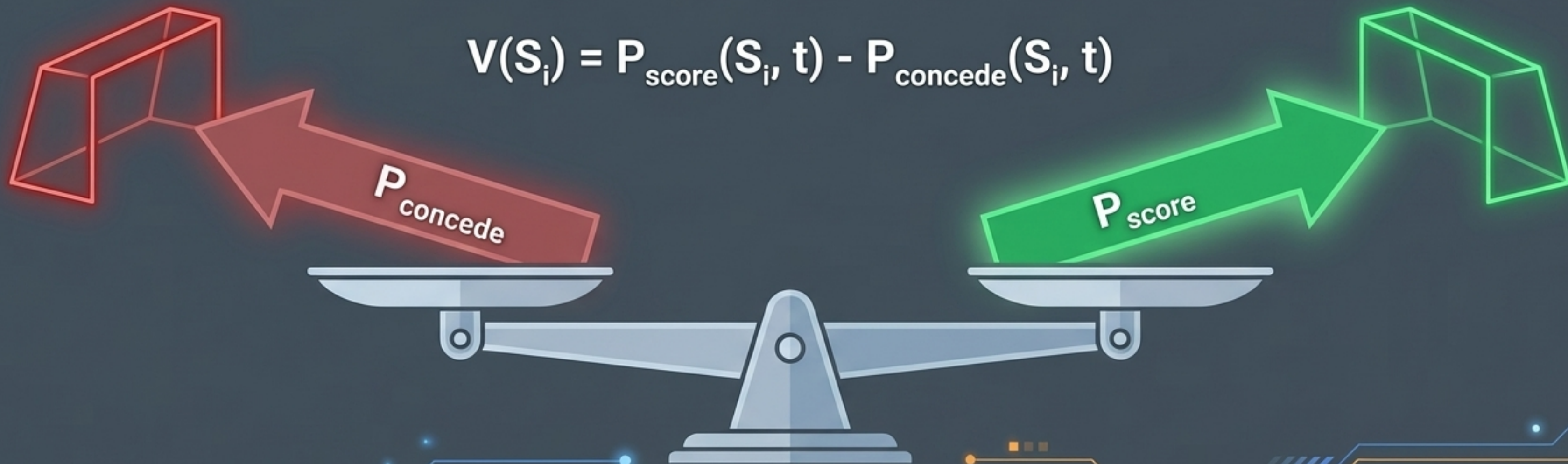
Researchers: Javier Fernandez et al. (2019)

Innovation: Upgraded from static event data to dynamic tracking data. Game states are modeled continuously via Deep Learning, factoring in the positioning, velocity, and spatial influence of all 22 players.

Significance: Solves the discrete zone limitation by modeling the true, complex fluidity of soccer space.

Quantifying the Risk-Reward Trade-off

$$V(S_i) = P_{\text{score}}(S_i, t) - P_{\text{concede}}(S_i, t)$$



Researchers

Tom Decroos et al. (2019)
VAEP Framework

Innovation

Actions are not exclusively offensive. VAEP values a game state by subtracting the probability of conceding from the probability of scoring in the next 10 actions.

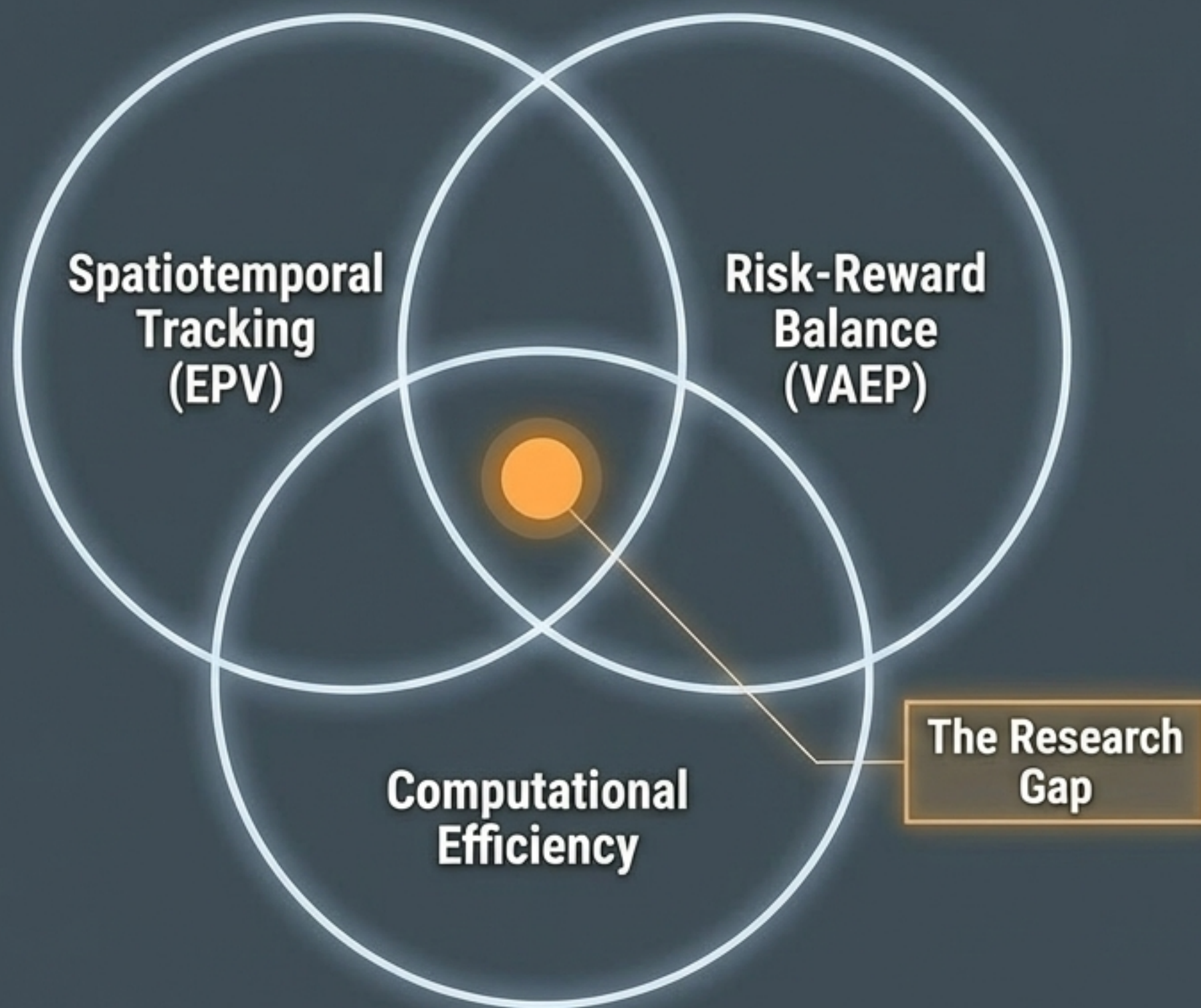
Significance

By utilizing Gradient Boosting on complex contextual features, VAEP successfully captures the risk of backward passes, turnovers, and the value of defensive actions.

Comparative Synthesis of State Value Frameworks

Model & Year	Primary Data Source	Spatial Resolution	Output Scope	Core Algorithm
Rudd (2011)	Event Data	Discrete Broad Zones	Offensive Reward	Markov Decision Process
Singh (2018 - xT)	Event Data	M x N Grid	Offensive Reward	Iterative / Linear Algebra
Fernandez (2019 - EPV)	Tracking Data	Continuous Spatiotemporal	Offensive Reward	Deep Learning Neural Nets
Decroos (2019 - VAEP)	Complex Event Data	Continuous Contextual	Risk & Reward	Gradient Boosting

Defining the Next Analytical Frontier



The Consensus

Valuing the 99% requires quantifying how deliberate actions alter state probabilities.

The Current Limitation

Existing tracking models (EPV) are computationally heavy and often ignore defensive risk, while risk-aware models (VAEP) rely on lower-fidelity event data.

Proposed Methodology

Developing a computationally efficient, tracking-based risk-reward model for off-ball defensive valuation.

Selected Foundational Literature

Rudd, S. (2011). A framework for tactical analysis and individual offensive production assessment in soccer using Markov chains. New England Symposium on Statistics in Sports.

Singh, K. (2018). Introducing Expected Threat (xT).

Fernandez, J., Bornn, L., & Cervone, D. (2019). Decomposing the immeasurable sport: A deep learning expected possession value framework for soccer. 13th MIT Sloan Sports Analytics Conference.

Decroos, T., Bransen, L., Van Haaren, J., & Davis, J. (2019). Actions Speak Louder than Goals: Valuing Player Actions in Soccer. KDD.